

SOFTWARE ENGINEERING PROJECT

Welcome to the Presentation of Our Project

Intelligent House System for Smart Lighting
and AI-Powered Security Monitoring

VIGILIS





PROJECT PRESENTATION

THEME:

Design and Implementation of an Intelligent House System Based on Light Intensity Control and Security Monitoring.

Software Engineering Project

In view of obtaining a **Higher Technician Diploma (HTD)** in Computer Sciences.

Option: Software Engineering

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Plan of Work

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Introduction

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Context & Problematic

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Conclusion & Perspectives

01 — INTRODUCTION

From Manual Homes to Intelligent Living



02

CONTEXT & PROBLEMATIC

From manual control to an intelligent, proactive home.

Where Today's Homes Fall Short



Security

- Manual monitoring
- Delayed detection
- No intelligent verification



Energy

- Manual lighting
- Unnecessary consumption
- No automatic brightness



Access

- Fragmented devices
- Limited remote control
- No unified platform



How can a residential environment improve security, lighting efficiency and emergency response by integrating IoT devices, AI-powered threat analysis, facial recognition and a centralized mobile application?

VIGILIS — Intelligent House System



Sense



Analyze



Decide



Respond

Sense → Analyze → Decide → Respond

Core Functionalities



Smart Lighting

Automatic ON/OFF and brightness adjustment



Facial Recognition

Identify or verify authorized persons



Motion Detection

Detect movement through connected sensors



AI Threat Analysis

Evaluate events and identify genuine threats



Remote Monitoring

View security status, cameras and events



Emergency Response

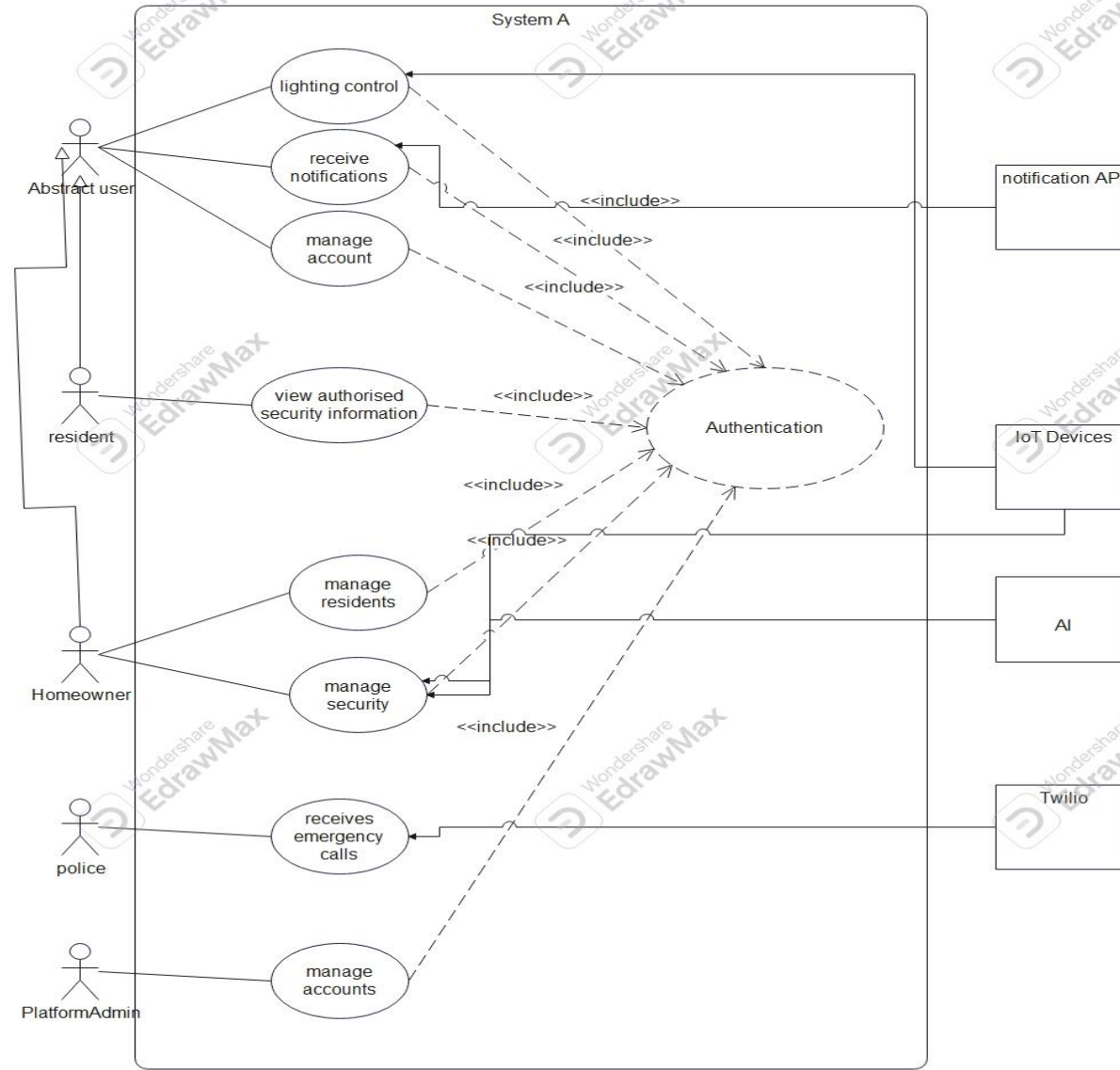
Notify the appropriate security service when confirmed

Intelligent Security Workflow

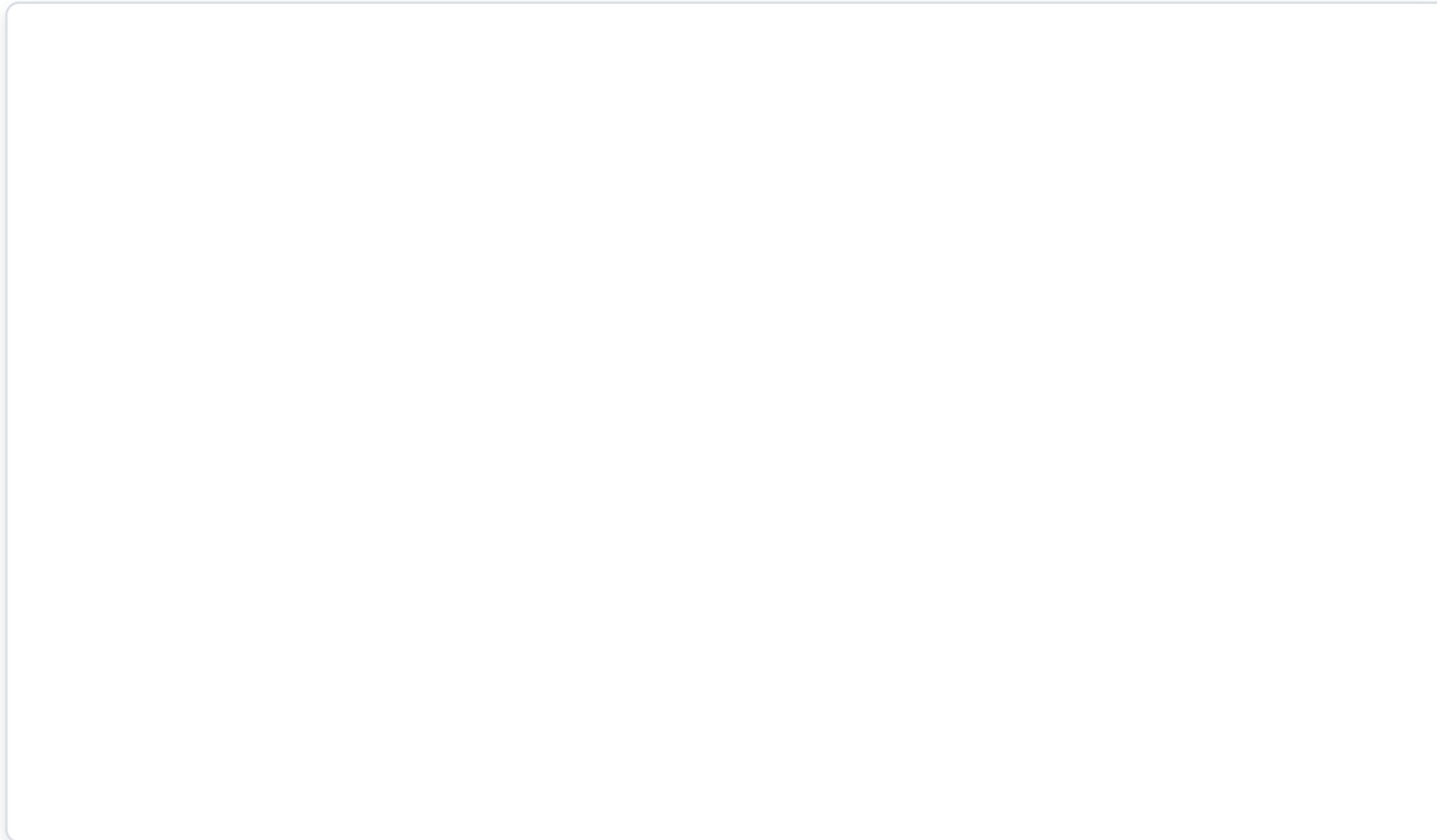


The AI acts only after the system has enough evidence to classify the event as a genuine threat.

General Use Case



Class Diagram



Design Stage

Preliminary design

Detailed design

Realisation branch

*Main domain entities and
relationships.*

Technology Stack



Flutter / Dart

Mobile frontend



Node.js + Express.js

REST API backend



Sequelize

ORM / database access



MySQL + XAMPP

Relational database



ESP32 / ESP32-CAM

IoT controller + camera

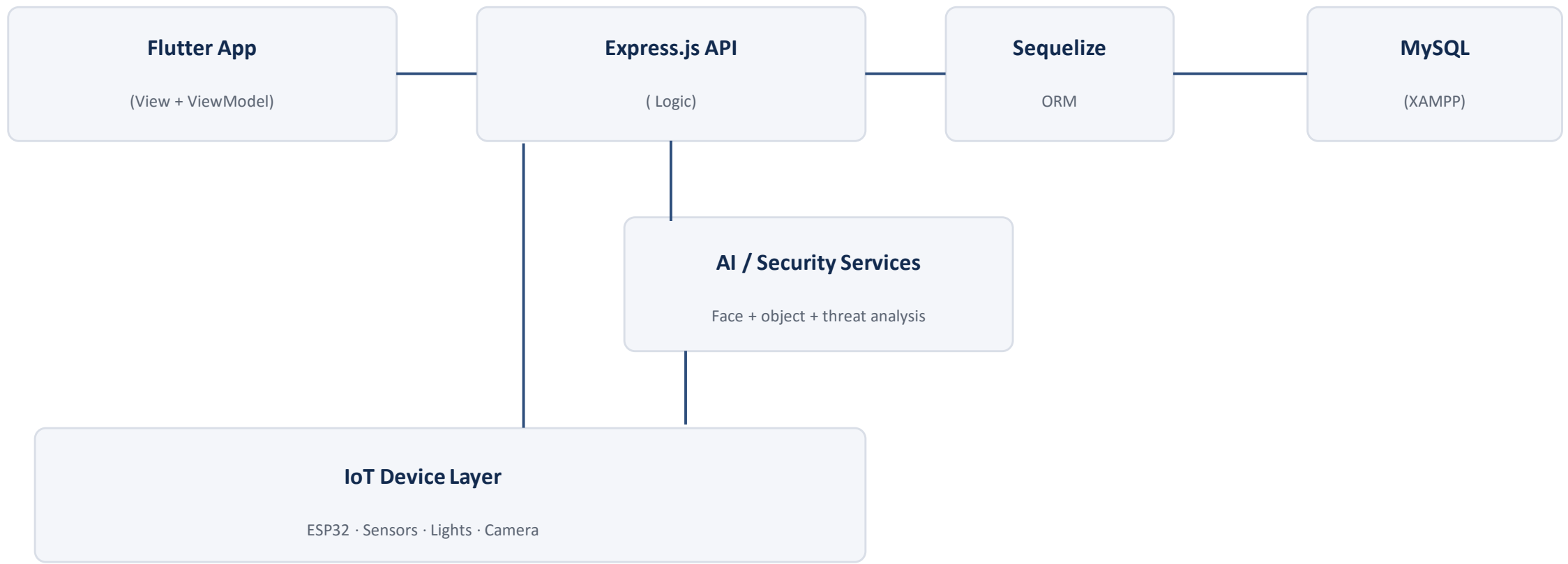


AI Services

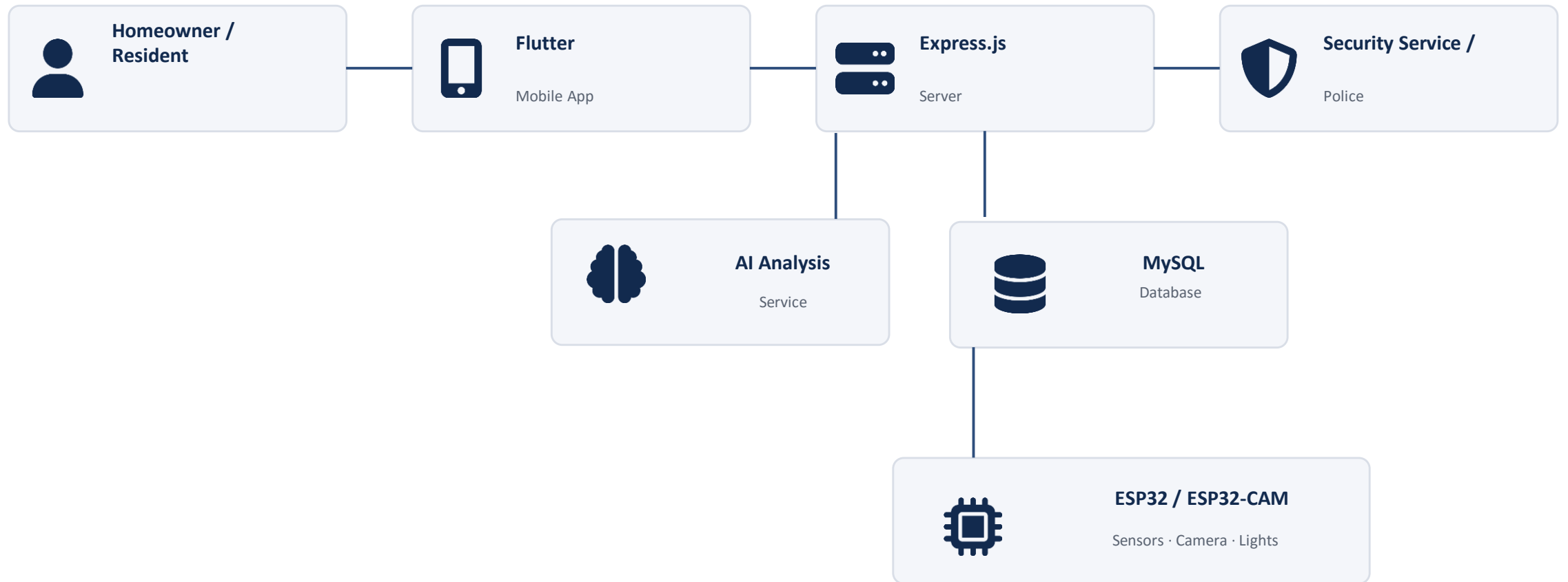
Face + object + threat analysis

Logical Architecture

MVVM + REST API + ORM + IoT / AI Integration



Physical Architecture



Each house is isolated by house ID / authorization rules to prevent cross-house device access.

Demo Flow

1



Login

2



Add Resident

3



Activate Security

4



Adjust Brightness

A house that senses.

A system that understands.

A response that acts.

VIGILIS brings together IoT, AI, smart lighting, secure access and emergency response to create a safer, smarter and more responsive home.